



Homework 12
—
GMM & EM



Basics

In the upcoming lecture, we will be covering the Expectation Maximization Algorithm (EM-Algorithm) for clustering Gaussian Mixture Models (GMM).

In this homework, you will first recapitulate and deepen your understanding of One-Dimensional Gaussian probability distributions – this should have been covered in your basic Math classes.

After this Recap, the two main videos of this homework introduce you to the Multidimensional Gaussian Distribution.



Recap – One-Dimensional Gaussian Distribution

Estimated time: 22min to watch the videos

1. one-dimensional Gaussian (Normal) Distribution

<https://www.youtube.com/watch?v=iYiOVISWXS4>

<https://www.youtube.com/watch?v=4R8xm19DmPM>

https://www.youtube.com/watch?v=-Uljlcq_rfc

These three videos introduce the one-dimensional Gaussian distribution starting from scratch. Again, I hope you already know all of this material. (*total time to watch: 22min*)



Video 1 – Multidimensional Gaussian Distribution

Estimated time: 15min to watch the video + 15min for the questions

♦ **Watch the video: “multivariate Gaussian (Normal) Distribution”**

<https://www.youtube.com/watch?v=eho8xH3E6mE>

This video also extends the 1-dim Gaussian distribution to the multi-dimensional case (called multivariate Gaussian distribution), but gives you more of a mathematical background (to understand it, you will need material covered in the Linear-Algebra Recap at the beginning of this homework, if you have forgotten about Eigenvalues, Rotation Matrices etc.)

♦ **Answer the following questions**

- 1) Why does the formula of the multivariate Gaussian Distribution have a constant factor in front of the “exp” function?
- 2) If you have 2 independent 1-dim Gaussian variables with their respective means and variances, what does the 2-dim Gaussian consisting of the them look like? What are it's means and covariance matrix?
- 3) Why would you want to decompose the covariance matrix of a multi-dim Gaussian into it Eigendecomposition?
- 4) What to the Eigenvectors of the covariance matrix correspond to?
- 5) What do the Eigenvalues of the covariance matrix correspond to?

To verify your answers: 1) to make it integrate to 1; 2) see timeindex 4:11ff in the video; 3) see video around timeindex 12:30; 4) the axes of the constant-probability ellipse; 5) The “length” of the constant-probability ellipse in the direction of the Eigenvectors;

- 1) ensures that it integrates to 1
- 2) joint distribution is product of individual for two variables x_1 and x_2
 x , mean and variance are summarised to concatenated x , mean and diagonal matrix (scaling)
- 3) generates collection of samples drawn from multivariate gaussian with arbitrary covariance matrix by decomposing that matrix
 understand geometry and role of eigenvectors and values in shape of ellipse
- 4) correspond to the axis of the constant probability ellipse
- 5) with eigenvalues as their length



$$p(x_1) = \frac{1}{Z} \exp \left\{ -\frac{1}{2\sigma_1^2} (x_1 - \mu_1)^2 \right\}$$

$$p(x_2) = \frac{1}{Z_2} \exp \left\{ -\frac{1}{2\sigma_2^2} (x_2 - \mu_2)^2 \right\}$$

$$\underline{x} = [x_1 \ x_2]$$

$$p(x_1)p(x_2) = \frac{1}{Z_1 Z_2} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\}$$

Congratulations $\underline{\mu} = [\mu_1 \ \mu_2]$

You are ready $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$

